Supplementary material 1. Search strategy

Pubmed	(**lung neoplasms**[Mesh] OR **lung cance***[tiab] OR **lung neoplas***[tiab]) AND ("**Programmed Cell Death 1 Receptor/antagonists and inhibitors**"[Mesh] OR **PD-1**[tiab] OR **PD1**[tiab] OR **PD-L1**[tiab] OR **CTLA4**[tiab] OR **CTLA-4**[tiab] OR **cytotoxic T-lymphocyte-associated protein 4**[tiab] OR **PDL1**[tiab] OR **Programmed Cell Death 1***[tiab] OR **Immune Checkpoint***[tiab] OR **nivolumab**[tiab] OR **pembrolizumab**[tiab] OR **atezolizumab**[tiab] OR **durvalumab**[tiab] OR **cemiplimab**[tiab] OR **avelumab**[tiab] OR **monoclonal antibodies** [tiab] OR **ipilimumab**[tiab] OR **tremelimumab**[tiab]) AND (**drug related side effects and adverse reactions**[Mesh] OR **adverse reactio***[tiab] OR **adverse even*[**tiab] OR **side effec***[tiab]) AND (**risk factor**[Mesh] OR **risk facto***[tiab] OR **predicto***[tiab] OR **exposur***[tiab]) AND ("2000/01/01"[Date - Publication] : "2020/11/12"[Date - Publication])
Embase	('**lung tumor**'/exp OR '**lung tumor**':ab,ti OR '**lung cancer**':ab,ti OR '**lung neoplasm***':ab,ti) AND (**'pdl1 gene'**/exp OR **'immune checkpoint inhibitor'**/exp OR **'immunomodulating agent'**/exp OR '**immunomodulating agent**':ab,ti OR '**avelumab**':ab,ti OR '**cemiplimab**':ab,ti OR '**durvalumab**':ab,ti OR '**atezolizumab**':ab,ti OR '**pembrolizumab**':ab,ti OR '**nivolumab**':ab,ti OR '**pdl1**':ab,ti OR '**pd1**':ab,ti OR '**immune checkpoint inhibitor**':ab,ti OR '**tremelimumab**':ab,ti OR ' **ipilimumab**':ab,ti OR **'ctla4'**:ab,ti OR **'ctla-4'**:ab,ti OR '**cytotoxic T-lymphocyte-associated protein 4**':ab,ti) AND ('**side effect**'/exp OR '**side effect**':ab,ti OR '**adverse drug reaction**'/exp OR '**adverse drug reaction**':ab,ti OR '**adverse event**'/exp OR '**adverse event**':ab,ti) AND ('**risk factor**'/exp OR '**risk factor**':ab,ti OR **'exposure'**/exp OR **exposure**:ab,ti OR **predicto***:ab,ti) AND [article]/lim AND [embase]/lim NOT ([embase]/lim AND [medline]/lim) AND [2000-2020]/py
Web of science	TS=("**lung neoplasms**" OR "**lung cance*"** OR "**lung neoplas***") AND TS= ("**Immune Checkpoint Inhibit*"** OR "**PD-1**" OR "**PD-L1**" OR "**PD1**" OR "**PDL1**" OR "**avelumab**" OR "**cemiplimab**" OR "**durvalumab**" OR "**atezolizumab**" OR "**pembrolizumab**" OR "**nivolumab**" OR "**CTLA-4**" OR "**CTLA4**" OR "'**tremelimumab**" OR " **ipilimumab**" OR " **cytotoxic T-lymphocyte-associated protein 4**") AND TS= ("**drug related side effects**" OR **"adverse reactio*"** OR "**adverse even*"** OR "**side effec***") AND TS= ("**risk facto*"** OR "**predicto*"** OR "**exposur***")